

Yulei Fu

US Citizen | yulei-machine-shop.com | yuleifu@umich.edu | linkedin.com/in/yulei-fu | 734-239-4771

EDUCATION

University of Michigan

Bachelors of Science in Robotics Engineering, GPA 3.21

Sep 2023 - May 2027

Ann Arbor, MI

SKILLS

Languages: C, C++, Rust, Python, Bash, Verilog

Software Tools: Embedded Linux, IPC, MQTT, GDB debugging, Docker, Google Test, CMake

Robotics/Autonomy: ROS2, Nav2, Google Cartographer, Slam Toolbox, Cyclone DDS

Relevant Coursework: Robot Perception and Computer Vision, Mobile Robotics (SLAM), Robot Operating Systems (ROS2), Autonomous Vehicles (Planning & Control), Data Structures & Algorithms, Computer Organization, Embedded Systems

EXPERIENCE

H3D, Inc.

June 2026 - August 2026

Robotics Engineering Intern

Ann Arbor, MI

- Owned an autonomous radiation-mapping UGV project for nuclear plant surveys to reduce human exposure to radioactive medium on a federal SBIR Phase I program, securing a Phase II award exceeding \$1M as of July 2026.
- Built and integrated full sensor stack using NVIDIA Jetson AGX Orin, ConnectTech Rogue carrier dual 10GbE, fusing Ouster OS0-128 LiDAR over ethernet, RealSense D455 depth camera, STM32F103RC ackermann-drive base.
- Deployed Nav2 autonomous navigation with waypoint following on the UGV, tuning global and local planner parameters (costmap inflation, controller lookahead, goal tolerances) to achieve 95% waypoint success over 80 runs in cluttered indoor environments while cutting recovery behavior triggers by 97%.

Boeing

Jan 2026 - June 2026

Simulation Research Intern

Hybrid

- Built IsaacSim digital twin models in Docker to virtually commission 3 robotic agents (HSR, KMR, RS007) in 2 human-in-the-loop manufacturing configurations to determine feasibility, cutting evaluation time by 85% (168 hrs to 15 hrs).
- Developed a CAD-to-URDF-to-USD asset conversion and scene composition pipeline for dynamic environment generation in IsaacSim, enabling reproducible reconfiguration across varying part geometries using ROS2 Jazzy.

Amazon Robotics

Jan 2025 - June 2025

Embedded Software Engineer Co-Op

Westborough, MA

- Delivered a C++ telemetry and log-capture pipeline for a safety-certified embedded Linux controller, publishing 10K+ system metadata messages/day to AWS IoT Core over MQTT for data persistence in S3.
- Reduced team Jira ticket backlog by 94% by saving metadata issue reproduction time by 85% (6 hrs to 45 min), enabling a 7-engineer diagnostics team to finally access consistent, fleet-reliable field data.
- Deployed and designed a remote Wi-Fi diagnostics API, adding to the system's hardware-diagnostics suite, reducing field-troubleshooting from 2+ hours to 0.03 hours for on-sites that previously had no remote pathway to inspect connectivity issues.

Stirling Research Group

Nov 2024 - Jan 2025

EVA Research Intern

Ann Arbor, MI

- Implemented the Keytel metabolic-rate model in JavaScript on a nRF52840 BLE-based heart-rate wearable to convert real-time heart-rate telemetry into energy-expenditure estimates, enabling accurate British Thermal Unit (BTU) tracking.

PROJECTS

MBot Planning and Perception (ROS2, C++)

- Implemented A* planning on a BFS obstacle-distance grid with line-of-sight path pruning, cutting waypoint count and producing 97% collision-free paths with clearance margins instead of wall-hugging grid staircases.
- Wrote the full SLAM pipeline by hand: a 400-particle filter with low-variance resampling and a log-odds occupancy grid at 5 cm resolution, subsampling every 5th lidar ray to sustain real-time localization on a Raspberry Pi.

LEADERSHIP

- Lead Instructional Assistant**, ROB 201 with Professor Jessy Grizzle)
- VP Career Development**, Robotics Undergraduate Student Council

Aug 2024 - Present

Aug 2026 - Present